Frequency Response (for LTI systems)

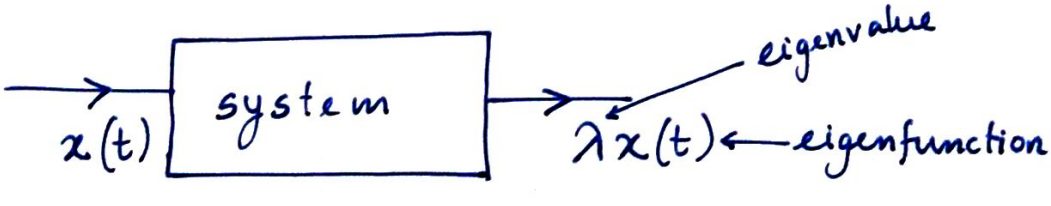
Frequency Response is a measure of the magnitude and phase of the output as a function of frequency. In order to find the frequency response we need to give as input a sine/cosine signal to the linear time-invariant system. There are many methods to find the output:

* Solve differential equation
  - find particular solution taking $\cos \omega_0 t$ as input
* Using impulse response
  - Convolve the impulse response with $\cos \omega_0 t$

The other method is that of eigenfunction.

Eigenfunction:

If the output signal is a scalar multiple of the input signal, we refer to the signal as an eigenfunction and the multiplier as the eigenvalue.



* Consider the system described by —

$$\frac{dy(t)}{dt} + 2y(t) = x(t) \longrightarrow (1)$$

Determine if each of the following functions is an eigenfunction of this system. If it is, find its eigenvalue.

(a) e^{-t} for all time

Let $y(t) = \lambda e^{-t}$ and given $x(t) = e^{-t}$

$\therefore$ From equation (1),

$$\frac{d}{dt}(\lambda e^{-t}) + 2\lambda e^{-t} = e^{-t}$$

$$\Rightarrow -\lambda e^{-t} + 2\lambda e^{-t} = e^{-t}$$

$$\therefore \quad \lambda e^{-t} = e^{-t} \Rightarrow \lambda = 1$$

$\therefore$ e^{-t} is an eigenfunction with eigenvalue 1.

(b) e^{jt} for all time.

Let $y(t) = \lambda e^{jt}$ and $x(t) = e^{jt}$

$$\therefore \frac{d}{dt}(\lambda e^{jt}) + 2\lambda e^{jt} = e^{jt}$$

$$\Rightarrow j\lambda e^{jt} + 2\lambda e^{jt} = e^{jt}$$

$$\Rightarrow (2+j)\lambda = 1 \Rightarrow \lambda = \frac{1}{2+j}$$

$\therefore$ e^{jt} is an eigenfunction and has eigenvalue $1/(2+j)$.

(c) $\cos t$ for all time.

Let $y(t) = \lambda \cos t$ and $x(t) = \cos t$

$$\frac{d}{dt}(\lambda \cos t) + 2\lambda \cos t = \cos t$$

$$\Rightarrow \lambda \sin t + 2\lambda \cos t = \cos t$$

$\therefore \nexists\, \lambda$ such that $\cos t$ is an eigenfunction.

(d) $u(t)$ for all time

Let $y(t) = \lambda u(t)$ and $x(t) = u(t)$

$$\frac{d}{dt} y(t) + 2y(t) = x(t) \Rightarrow \lambda \delta(t) + 2\lambda u(t) = u(t)$$

$\therefore$ There exist no λ such that $u(t)$ is a eigenfunction.

From the above we have an intution that complex exponentials are important.

$\boxed{}$ All complex exponentials are eigenfunctions.

<u>Proof:</u> Let $x(t) = e^{st}$ where $s \in \mathbb{C}$

$\therefore y(t) = h(t) * x(t)$, where $h(t)$ is the impulse function of the system.

$$= \int_{-\infty}^{\infty} h(\tau) x(t-\tau)\, d\tau$$

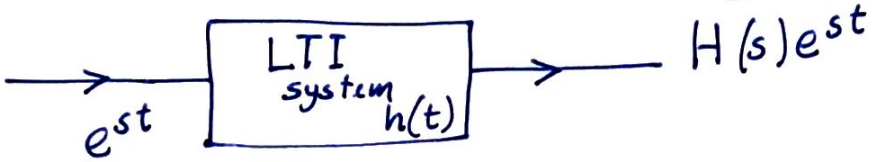
$$= \int_{-\infty}^{\infty} h(\tau) e^{s(t-\tau)}\, d\tau$$

$$\therefore \quad y(t) = e^{st} \underbrace{\int_{-\infty}^{\infty} h(\tau)\, e^{-s\tau}\, d\tau}_{\text{System function (Transfer function) evaluated at } s}$$

$$\therefore \quad y(t) = H(s)\, e^{st} \longrightarrow (2)$$

eigenvalue associated with e^{st}.

$\therefore$



Let us use an example to see how we can compute the frequency response of a system.

e.g. Find the response of the system described by $\frac{d}{dt} y(t) + 2\, y(t) = x(t)$ to the input $x(t) = e^{2jt}$ for all time.

<u>Solution:</u> The transfer function of the system is:

$$sY(s) + 2Y(s) = X(s)$$

$$\Rightarrow \quad H(s) = Y(s)/X(s) = \frac{1}{s+2}$$

$\therefore$ The output due to input $x(t) = e^{2jt}$ is

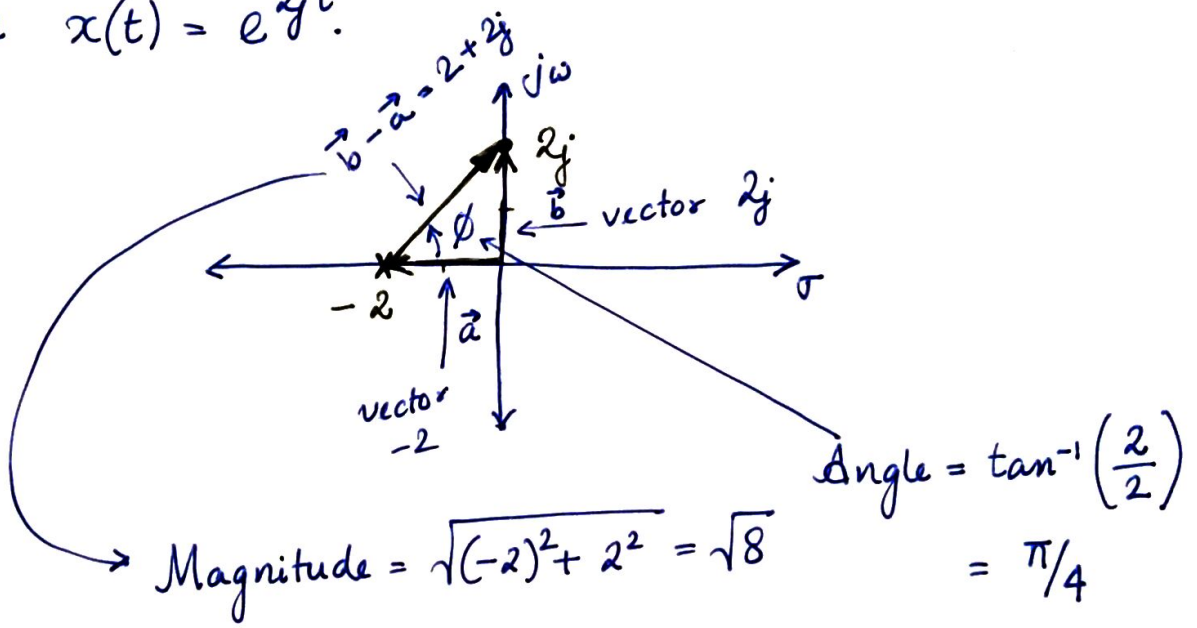
$$y(t) = H(2j)e^{2jt} \quad (\text{see eq}^n \text{ ②})$$

$$= \frac{1}{2j+2} e^{2jt}$$

<u>Graphical method to write/evaluate</u> $\left(\frac{1}{2+2j}\right)$.

$H(s) = \frac{1}{s+2} \Rightarrow$ Pole at $s = -2$

An we want to find the response at $2j$ $(\omega = 2)$ since $x(t) = e^{2jt}$.



$$\therefore \frac{1}{2+2j} = \frac{1}{\sqrt{8}e^{j\pi/4}} = \frac{1}{\sqrt{8}} e^{-j\pi/4}$$

$$\therefore \boxed{y(t) = \frac{1}{\sqrt{8}} e^{-j\pi/4} \cdot e^{2jt}}$$

Response due to a cosine input:

The input $x(t) = \cos \omega_0 t = \frac{1}{2}\left(e^{j\omega_0 t} + e^{-j\omega_0 t}\right)$

$\because$ The system is linear time-invariant, we have output

$$y(t) = \frac{1}{2}\left(H(j\omega_0)e^{j\omega_0 t} + H(-j\omega_0)e^{-j\omega_0 t}\right) \longrightarrow (3)$$

Recall that the system function is defined as

$$H(s) = \int_{-\infty}^{\infty} h(t)e^{-st}dt,$$ where $h(t)$ is the impulse response of the system.

$$\therefore H(j\omega_0) = \int_{-\infty}^{\infty} h(t)e^{-j\omega_0 t}dt$$

$$H(-j\omega_0) = \int_{-\infty}^{\infty} h(t)e^{j\omega_0 t}dt$$

For practical systems, the impulse response $h(t)$ is real valued. Using this fact we get

$$\therefore (H(j\omega_0))^* = \int_{-\infty}^{\infty} (h(t))^* e^{j\omega_0 t}dt = \int_{-\infty}^{\infty} h(t)e^{j\omega_0 t}dt$$

$$= H(-j\omega_0) \longrightarrow (4)$$

Using equation (4) in equation (3),

$$y(t) = \frac{1}{2}\left[H(j\omega_0)\, e^{j\omega_0 t} + H(j\omega_0)^* e^{-j\omega_0 t}\right]$$

$$= \frac{1}{2}\left[H(j\omega_0)\, e^{j\omega_0 t} + \left\{H(j\omega_0)\, e^{j\omega_0 t}\right\}^*\right]$$

$$= \frac{1}{2} \times 2 \; Re\left\{H(j\omega_0)\, e^{j\omega_0 t}\right\}$$

$$= Re\left\{|H(j\omega_0)|\, e^{\angle H(j\omega_0)}\, e^{j\omega_0 t}\right\}$$

$$= |H(j\omega_0)|\; Re\left\{e^{\angle H(j\omega_0) + j\omega_0 t}\right\}$$

$$\therefore \boxed{y(t) = |H(j\omega_0)| \cos\left(\omega_0 t + \angle H(j\omega_0)\right)} \longrightarrow (5)$$

e.g. Let us find the output of the system with transfer function $H(s) = \frac{1}{s+2}$ due to input $x(t) = \cos(2t)$.

<u>Solution:</u> Here $\omega_0 = 2$. $H(j2) = \frac{1}{2+2j}$

$\therefore |H(2j)| = \frac{1}{\sqrt{8}}$ and $\angle H(2j) = -\pi/4$

$\therefore$ From equation ⑤,

$$\boxed{y(t) = 1/\sqrt{8} \cos\left(2t - \pi/4\right)}$$ ← Frequency response at $\omega = 2$ rad/s.

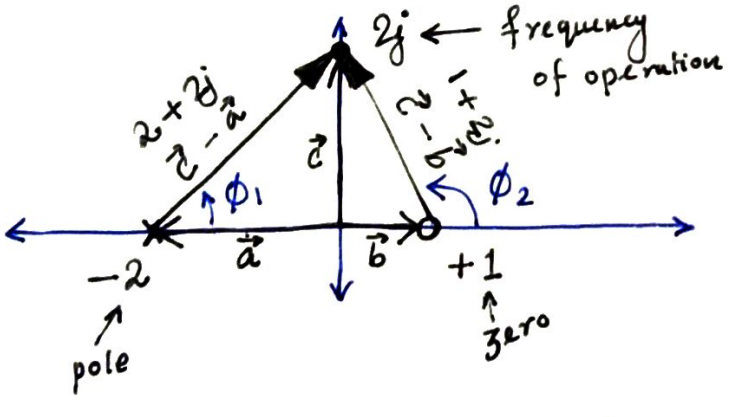
Let us look into another example that has a zero as well.

e.g Find the output of the system with transfer function $\frac{s-1}{s+2}$ due to input $x(t) = e^{2jt}$.

<u>Solution:</u> The output is

$$y(t) = H(2j)\, e^{2jt} = \underbrace{\left(\frac{-1+2j}{2+2j}\right)}\, e^{2jt}$$

We evaluate this graphically. This can be done by rationalization also.



$$\therefore y(t) = \frac{1/\sqrt{5}\, e^{j\phi_2}}{1/\sqrt{8}\, e^{j\phi_1}}\, e^{2jt} = \frac{\sqrt{8}}{\sqrt{5}}\, e^{j(\phi_2 - \phi_1)}\, e^{j2t},$$

$\phi_1 = \pi/4$, $\phi_2 = \pi - \tan^{-1}(2)$.

Continuous time Fourier series and LTI systems

Given a periodic signal $x(t)$ with time period T, its fourier series representation is —

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}, \quad \text{where } a_k = \frac{1}{T}\int_T x(t) e^{-jk\omega_0 t} dt$$

If such a signal is input to a LTI system with transfer function $H(s)$, then the output is

$$\boxed{y(t) = \sum_{k=-\infty}^{\infty} a_k H(jk\omega_0) e^{jk\omega_0 t}} \longrightarrow (6)$$

Note that $y(t)$ is also a periodic signal with fourier series representation given by equation (6).

The spectral co-efficients are :

$b_k = a_k H(jk\omega_0)$, where a_k is the spectral co-efficient of the kth harmonic of $x(t)$.

Q: What will be the output of a system with $H(s) = \frac{1}{s+2}$ if input is a square-wave with period 1? } Problem for Practice

Discrete-time case: Till now we have been looking into frequency response of LTI systems in the continuous-time case. The results for CT case analogously follows for discrete-time case.

All complex exponentials are eigenfunctions.

Let $x[n] = e^{j\Omega_0 n}$ for all n.

$\therefore$ The output of the LTI system with impulse response $h[n]$ is:

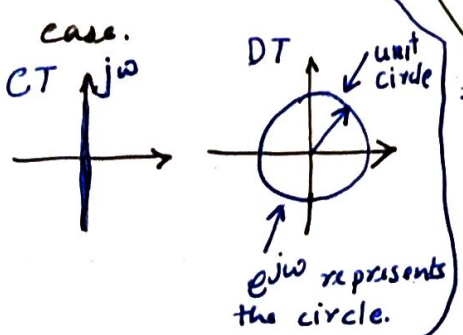
$$y[n] = \sum_{m=-\infty}^{\infty} h[m]\, x[n-m]$$

$$= \sum_{m=-\infty}^{\infty} h[m]\, e^{j\Omega_0 (n-m)}$$

$$= \left(\sum_{m=-\infty}^{\infty} h[m]\, e^{-j\Omega_0 m} \right) e^{j\Omega_0 n}$$

$$= \underbrace{H(e^{j\Omega_0})}_{\text{System function evaluated at } z = e^{j\Omega_0}} e^{j\Omega_0 n}$$

Difference with CT: In CT we had $H(j\omega)$ and in DT $H(e^{j\omega})$. This is because $j\omega$ axis is analogous to a circle in DT case.

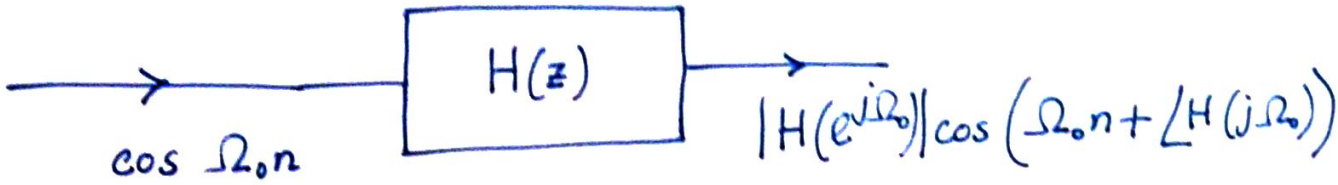


Recall:

$$H(z) = \sum_{m=-\infty}^{\infty} h[m]\, z^{-m}$$

$\therefore$ All the results of CT and DT are analogous in this case.

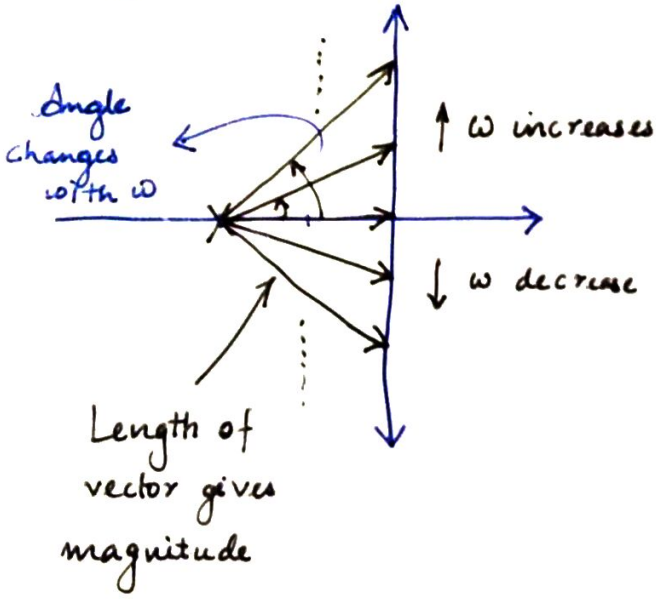
$\therefore$

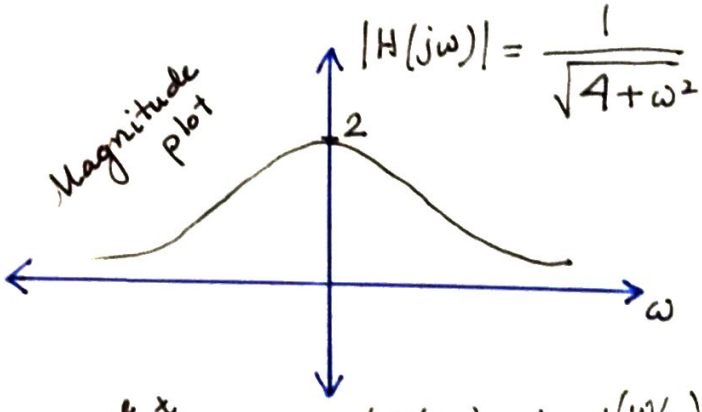


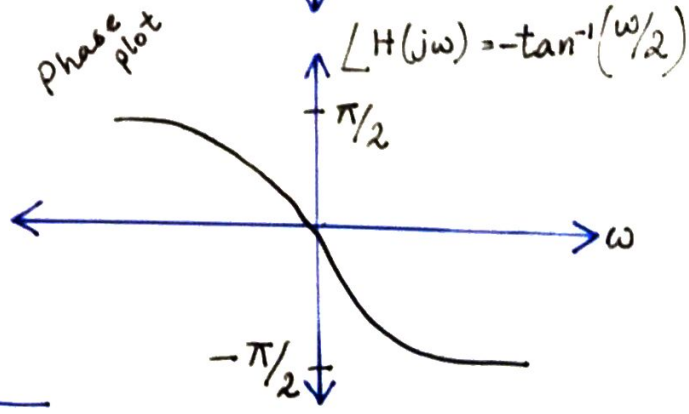
<u>Work for you</u>: Reproduce the results for the discrete as we did for the continuous case.

Till now the examples that we have looked into were all analysis at a particular frequency. Let us revisit the first example that we took and find the frequency response — basically we want to find the output as a function of frequency and plot it.

e.g. The transfer function is — $\frac{1}{s+2} = H(s)$







Since we have only one pole and no zero, the magnitude decreases with frequency.